

SpatialBlock: Enhancing Spatial Intelligence in LVLMs via Synthetic Block-Stacking Problem

Soohyun Ryu^{1*}, Sohee Kim^{1*}, and Eunho Yang^{1,2†}

¹ Korea Advanced Institute of Science and Technology (KAIST), South Korea

² AITRICS, South Korea

{rsoohyun, joyhee, yangeh}@kaist.ac.kr

Abstract. Large Vision-Language Models (LVLMs) have achieved strong performance on diverse visual tasks, yet their ability to reconstruct and reason about the 3D structure of the scene depicted in 2D images – referred to as *spatial intelligence* – remains limited. Existing approaches attempt to address this gap by using real-scene spatial question answering datasets that require dense geometric annotations. However, constructing such labels is costly, time-consuming, and often noisy due to reliance on external perception modules. In this work, we propose a novel paradigm inspired by human cognitive development: learning foundational spatial skills through structured block-manipulation tasks. We introduce **SpatialBlock-15k**, a synthetic dataset of 15,000 block-stacking problems covering 3D-to-2D projection, viewpoint transformation, and structural combination. The dataset further incorporates controlled color modulation as visual cues to encourage anchor-based reasoning in visually complex conditions. Experiments demonstrate that LVLMs trained on our dataset through either direct answering or reasoning-based prediction significantly outperform baselines and generalize to real-world spatial tasks, despite the dataset’s synthetic and compact nature.

Keywords: Spatial Intelligence · Large Vision-Language Models · Synthetic Dataset

1 Introduction

Large Vision–Language Models (LVLMs) [1, 22, 29] have demonstrated remarkable performance across a wide range of 2D image understanding and reasoning tasks. Despite this progress, their ability to mentally reconstruct the 3D structure of a scene from 2D images – namely, *spatial intelligence* – remains limited. This shortcoming poses a fundamental bottleneck for deploying LVLMs in real-world applications that require robust spatial reasoning, such as autonomous driving [20] and robotics [8].

To address this limitation, prior work has extended model architectures by adding a specialized module for learning 3D feature [6, 23] or designed tasks that

* Equal contribution, † Corresponding Author.

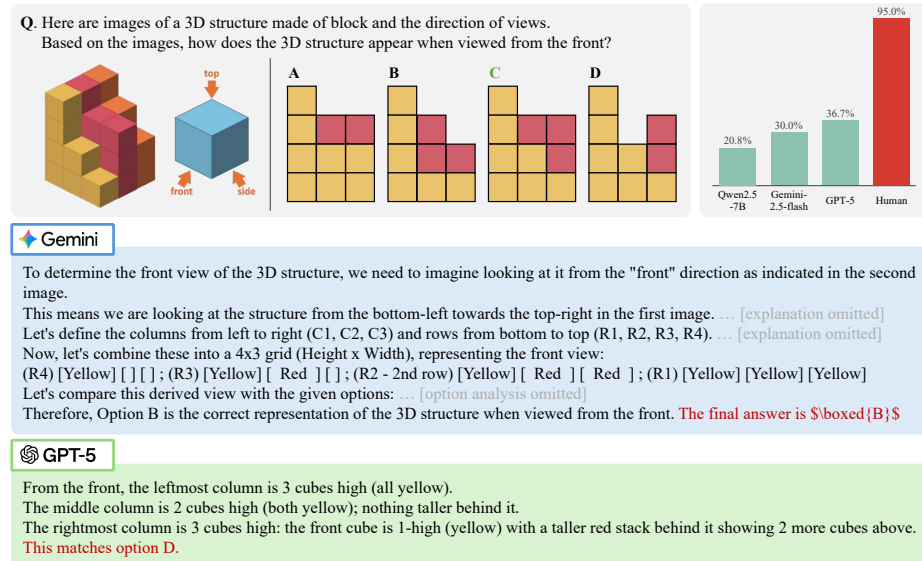


Fig. 1: Performance Gap Between Humans and LVLMs in Block-Stacking Problem. The top row illustrates an example block-stacking task that requires predicting a 2D projection from a given 3D structure. While humans achieve near-perfect accuracy, state-of-the-art LVLMs struggle even with basic spatial reasoning problems. Leading proprietary models (*e.g.*, Gemini and GPT-5) exhibit flawed spatial logic, often producing incorrect interpretations of the structure.

directly require spatial reasoning [24, 27] to encourage models to learn such capabilities. However, these approaches typically rely on real-world scene question answering datasets that require dense geometric and object-level annotations of the visual context. Constructing such dense labels is not only time-consuming and costly, but also noisy due to the frequent reliance on external modules (*e.g.*, segmentation or depth estimation models), thereby limiting their reliability and scalability.

In contrast to methods that depend on annotation-heavy real-scene data, we pursue a more efficient alternative inspired by the way humans develop spatial intelligence. Developmental psychology [10, 12] suggests that foundational spatial abilities are often cultivated through structured manipulative tools. Specifically, block play has been widely shown to enhance essential components of spatial cognition, including spatial integration and mental simulation [3, 4]. Despite its simplicity, state-of-the-art LVLMs struggle with such problems, suggesting that they lack even fundamental levels of spatial intelligence (Fig. 1).

Motivated by this observation, we propose **SpatialBlock-15k**, a novel synthetic dataset of block-stacking problems designed to improve spatial intelligence in LVLMs. The dataset comprises three categories of spatial reasoning tasks: (1) 3D-to-2D projection, (2) viewpoint transformation, and (3) structural combination. To better approximate real-world spatial reasoning scenarios, where

humans interpret scenes by anchoring on a specific object and reasoning relationships relative to it, we introduce color modulation as a visual cue, encouraging models to identify task-relevant elements in visually complex scenes. Unlike costly real-scene datasets, our dataset can be generated synthetically, enabling clean, scalable, and cost-effective data construction.

Experimental results show that models trained on SpatialBlock-15k, with either direct answer prediction or reasoning-based prediction, significantly outperform baseline models, even though the training data are entirely synthetic and relatively small in scale. These findings suggest that targeted training on fundamental spatial reasoning tasks can effectively enhance the spatial intelligence of LVLMs and provide a scalable alternative to annotation-heavy real-scene supervision.

In summary, our contribution is three-fold:

- We introduce a new perspective on enhancing spatial intelligence in LVLMs by shifting from annotation-heavy real-scene supervision to foundational spatial skill learning through structured synthetic tasks.
- We present **SpatialBlock-15k**, a scalable synthetic dataset of block-stacking problems spanning 3D-to-2D projection, viewpoint transformation, and structural combination, enhanced with controlled color cues for task-relevant reasoning.
- We show that training LVLMs on SpatialBlock-15k significantly improves spatial reasoning performance and generalizes effectively to real-world scenes, despite relying solely on synthetic data.

2 Related Work

Visual Spatial Reasoning in LVLMs Recent efforts to improve spatial intelligence in LVLMs focus on building a dedicated encoder or proposing a specialized training strategy. On the architectural side, some approaches modify the model by introducing spatial tokens into the vision encoder [15, 21], incorporating depth-aware plugin modules [6], or adding spatial encoders to learn 3D features [23]. Other research focuses on the training process, designing hierarchical schemes that transition from spatial perception to complex reasoning [14, 16] or encouraging the model to output cognitive maps that encode object position and orientation [24, 27]. Despite their effectiveness, these approaches generally require real-scene annotations or pseudo-3D signals from external modules, which limits their scalability and robustness.

Training Datasets for Spatial Intelligence Various training datasets have been proposed to enhance the spatial reasoning capabilities of LVLMs, ranging from simple spatial relations to complex multi-step reasoning. Early benchmarks [5, 17] leverage external models (*e.g.*, object detection, segmentation, depth, or pose estimation) to extract 3D information and generate low-level

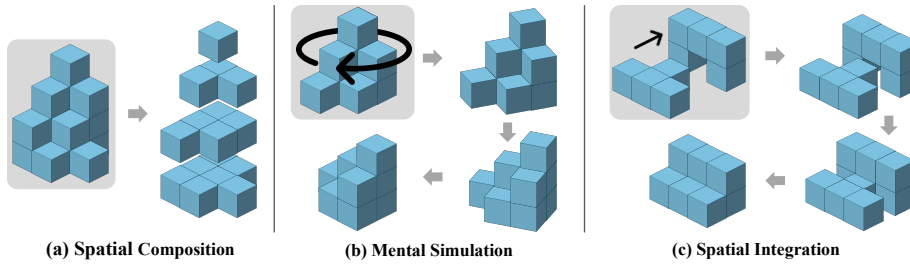


Fig. 2: Spatial Abilities Underlying Block-Stacking Reasoning. Solving block-stacking tasks involves (a) inferring latent 3D composition, (b) reasoning under transformation, and (c) integrating multiple components into a coherent structure.

QA pairs such as distance, orientation, and spatial relations. To foster higher-level logic, recent works [18, 24] use 3D annotated video data to construct relative relation and direction reasoning, or more sophisticated tasks such as route planning or spatio-temporal appearance ordering through human annotation. MindCube [27] focuses on spatial mental modeling, which requires synthesizing a full scene from partial views or inferring arrangements under perspective shifts. Despite these advancements, a significant bottleneck remains: these approaches heavily rely on dense, real-scene 3D annotations, which are labor-intensive and inherently noisy, necessitating costly human annotation.

3 SpatialBlock-15k

We introduce **SpatialBlock-15k**, a scalable synthetic dataset designed to systematically probe and enhance foundational spatial abilities, moving away from the conventional annotation-intensive paradigm that relies on dense real-scene 3D annotations. Specifically, we utilize block-stacking problems, which play a pivotal role in the early stages of human spatial cognitive development (Sec. 3.1). Furthermore, we incorporate controlled color variations as visual cues to better reflect real-world spatial reasoning conditions (Sec. 3.2).

3.1 Block-Stacking Problems for Spatial Intelligence

Activities involving block-stacking require structural spatial understanding that extends beyond superficial visual pattern recognition. As illustrated in Fig. 2, solving such problems requires (a) spatial composition, the ability to reconstruct the complete 3D structure by deconstructing the visible configuration and inferring occluded blocks that must exist to support it; (b) mental simulation, the capacity to simulate the outcome of transformations applied to a 3D structure; and (c) spatial integration, the ability to anticipate the resulting structure when multiple components are combined. Together, these abilities reflect core dimensions of spatial intelligence.

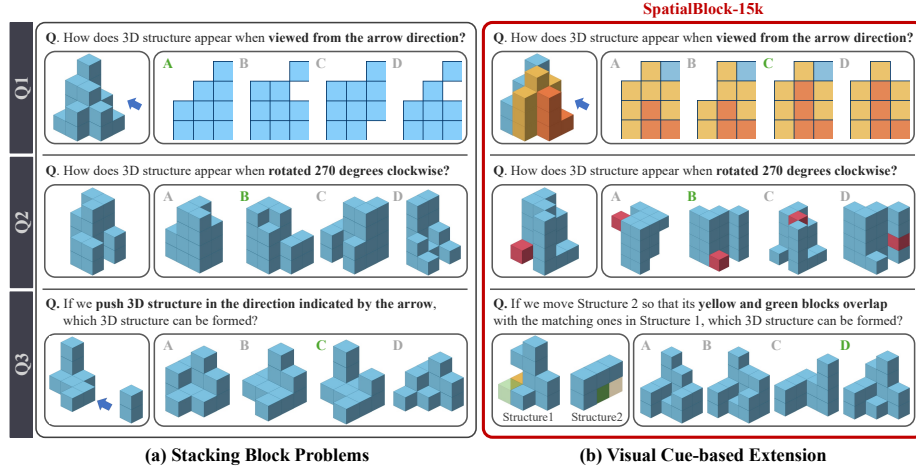


Fig. 3: SpatialBlock-15k. (a) Three block-stacking tasks: 3D-to-2D projection (Q1), viewpoint transformation (Q2), and structural combination (Q3). (b) Visual cue-based extension with controlled color variations, where colors encode depth ordering (Q1), preserve structural roles across transformations (Q2), and provide correspondence cues for integration (Q3).

Building on these three abilities, we define three question types, each targeting a distinct aspect of spatial reasoning (Fig. 3(a)):

- Q1: 3D-to-2D projection** This task requires predicting the 2D appearance of a 3D structure from a specific viewing direction. Solving it demands the ability to internally reconstruct the 3D configuration and mentally transform it to the target viewpoint. The model must additionally perform depth-aware occlusion reasoning to determine which blocks remain visible in the final 2D projection.
- Q2: Viewpoint transformation** The problem inquires how a given 3D structure appears under self-rotation or viewpoint changes. The model should preserve structural consistency across transformations, maintaining correct relative positions of all components without distortion, disappearance, or unintended displacement.
- Q3: Structural combination** This question asks how the resultant integration is formed when two 3D structures are combined. It requires simulating how individual components interact, identifying contact interfaces, and subsequently inferring the coherent global structure that emerges from their integration.

Collectively, these question types form the foundation of our SpatialBlock-15k dataset, providing a structured framework for evaluating and training spatial reasoning in LVLMS.

3.2 Visual Cue-based Extension

In the previous section, we constructed tasks that require understanding 3D structures composed of single-color blocks. Since color does not provide a discriminative feature, these tasks demand the model to focus on purely geometric relationships, such as the relative positions of blocks and appearance changes under viewpoint transformations.

However, real-world scenes typically consist of complex environments with diverse objects, rather than visually uniform ones. Given this intricacy, humans tend to analyze the global scene configurations by selecting a specific object as an anchor and interpreting relationships relative to it [11]. That is, visually salient cues serve as the reference points to infer changes and relations.

Motivated by this cognitive characteristic, we extend the monochromatic block-based tasks introduced in Sec. 3.1 by incorporating color as an additional cue. Here, color is not used merely for visual diversity but as a functional guidance that reveals depth information, structural correspondences, and contact regions. This design encourages the model to reconstruct the global structure and track transformations by using specific blocks as reference points. Specifically, we extend the three question types as follows (Fig. 3(b)):

- Q1: 3D-to-2D projection with depth cues** To facilitate depth-aware reasoning, different colors are assigned according to depth from a given viewpoint. As a result, the task requires not only matching the silhouette but also tracking front-back relationships using color information. Color functions as an explicit indicator of depth ordering and is designed to clearly reflect the hierarchical organization of the 3D structure.
- Q2: Viewpoint transformation with anchor blocks** Among the blocks composing the 3D structure, one anchor block is assigned a distinct color. This block must maintain the same structural role before and after transformation. Accordingly, the task requires not only recognizing geometric structure but also understanding anchor-relative relationships that are preserved through transformation.
- Q3: Structural combination with overlapping blocks** The attachment location between two structures is represented not by arrows but by the overlap of colored blocks indicated by transparent regions. Each structure is presented as a separate image, so aligning the positions of same-colored blocks across images is needed to predict the combined structure. Color thus acts as a shared reference point linking two independent inputs, inducing reasoning inter-structural integration.

This visual cue-based extension is designed to maintain the requirement for understanding the 3D structure itself while additionally enabling the learning of scene organization and relational reasoning centered around anchor objects, as required in realistic visual contexts. Consequently, it leads to meaningful improvement in real-scene spatial reasoning performance, as empirically demonstrated in Sec. 5.3. Finally, the SpatialBlock-15k dataset comprises three question types with visual cue-based extensions, each containing 5k samples. Further details on data construction can be found in Sec. A.

4 Method

To foster the spatial intelligence of LVLMs using our SpatialBlock-15k dataset, we propose two distinct training strategies. Each recipe is designed to target different aspects of the model’s capabilities: (1) direct answer prediction, which focuses on cultivating rapid inference by mapping visual inputs directly to their corresponding answers (Sec. 4.1), and (2) reasoning-based prediction, which encourages the model to explain its intermediate logical paths that lead to a final answer (Sec. 4.2).

4.1 Direct Answer Prediction Model

To maximize the model’s capacity for instantaneous spatial problem-solving, we train it with supervision only on the ground-truth answer sequence $y = \{y_1, y_2, \dots, y_T\}$. Specifically, given a textual query q and image v , we optimize the model parameters θ to predict the next token y_i , given the preceding context $y_{(1:i-1)}$, by minimizing the standard cross-entropy loss:

$$\mathcal{L}_{ce}(\theta) = - \sum_i \log P(y_i | y_{(1:i-1)}, q, v). \quad (1)$$

This strategy directly optimizes answer precision, thereby establishing a foundation for high-fidelity spatial problem-solving proficiency.

4.2 Reasoning-based Prediction Model

To go beyond simple answer prediction, we adopt a second strategy that enables the model to articulate its internal logic. To ensure a solid foundation for spatial intelligence, we first initialize the model through supervised fine-tuning to establish basic task-solving ability, then employ reinforcement learning to facilitate high-order structural reasoning.

For model initialization, we train the model with Low-Rank Adaptation (LoRA) instead of full-parameter fine-tuning, as it often degrades the inherent reasoning ability. This lightweight adaptation maintains the model’s Chain-of-Thought (CoT) reasoning capabilities while aligning it with our dataset. Notably, we deviate from the conventional “cold-start” phase that relies on synthesized trajectories from larger teacher models [9, 14]. The rationale behind this decision is that even state-of-the-art proprietary models exhibit relatively low accuracy on our dataset (Tab. 1), making it infeasible to construct reliable high-quality trajectories. Our empirical analysis (Sec. 5.3) demonstrates a clear performance advantage in preserving the model’s reasoning ability through LoRA-based tuning. This approach provides a more effective initialization for RL training than attempting to recover the thinking ability via a low-quality cold-start phase.

After LoRA-based initialization, we directly optimize the model using reinforcement Learning via Group Relative Policy Optimization (GRPO). We design

a multi-objective reward function that evaluates response correctness and reasoning trace quality with three components:

$$\mathcal{R}(y, o) = \mathcal{R}_{\text{acc}}(y, o) + \mathcal{R}_{\text{format}}(o) + \mathcal{R}_{\text{len}}(o), \quad (2)$$

where o is a model-generated prediction given a question $c = (q, v)$ with ground-truth answer y . The accuracy reward $\mathcal{R}_{\text{acc}}$ verifies whether the final answer extracted from o matches y .

Next, the format reward $\mathcal{R}_{\text{format}}$ enforces compliance with the required CoT structure, where the model must provide a comprehensive reasoning trace followed by the final answer encapsulated within `<answer>` and `</answer>` tags. Lastly, to prevent degenerative behaviors such as overly brief or excessively verbose output, a length reward $\mathcal{R}_{\text{len}}$ is applied, requiring the response length L to satisfy $50 < L < 1024$. For simplicity and stability, all the aforementioned components are implemented in binary rewards, assigning a discrete value of 1 or 0 based on the satisfaction of each condition.

Given this reward, for each question c , the old policy model $\pi_{\theta_{\text{old}}}$ samples a group of candidate responses $\{o_1, o_2, \dots, o_G\}$. Each response o_i receives a reward $r_i = \mathcal{R}(y, o_i)$, from which we compute the advantage $A_i = \frac{r_i - \text{mean}(r_1, r_2, \dots, r_G)}{\text{std}(r_1, r_2, \dots, r_G)}$. The model is then updated by maximizing the following objective:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}_{c, \{o_i\}} \left[\frac{1}{G} \sum_{i=1}^G \min \left(\frac{\pi_{\theta}(o_i|c)}{\pi_{\theta_{\text{old}}}(o_i|c)} A_i, \text{clip} \left(\frac{\pi_{\theta}(o_i|c)}{\pi_{\theta_{\text{old}}}(o_i|c)}, 1 \pm \varepsilon \right) A_i \right) - \beta \text{KL}[\pi_{\theta} || \pi_{\text{ref}}] \right], \quad (3)$$

where ε and β are hyper-parameters, and $\text{KL}[\pi_{\theta} || \pi_{\text{ref}}]$ is the KL divergence between the policy model π_{θ} and reference model π_{ref} .

5 Experiments

We conduct multiple experiments to validate the effectiveness of block-stacking problems in fostering spatial intelligence. We first describe our experimental setup, including training details and evaluation protocols (Sec. 5.1). We then compare our models with prior methods across several spatial reasoning benchmarks (Sec. 5.2). In addition, we perform ablation studies to examine the effects of visual cues and the model initialization for the reasoning model (Sec. 5.3). Lastly, we present a comprehensive analysis of our proposed dataset, covering reasoning quality, generalizability, and robustness (Sec. 5.4).

5.1 Experimental Settings

Implementation details We adopt Qwen2.5-VL-3B [2], Qwen2.5-VL-7B [2], and Qwen3-VL-4B [1] as baseline models for our framework. To distinguish the training strategy, we append a suffix to the model name: *SpatialBlock-direct* denotes the model that outputs the answer directly (Sec. 4.1), while *SpatialBlock-reason* refers to the model that performs reasoning before producing the final answer (Sec. 4.2). Both variants are trained using the entire SpatialBlock-15k dataset.

Table 1: Evaluation Results on Spatial Reasoning Benchmarks. For each dataset, **bold** indicates the best performance and underlined represents the second-best performance. * represents that the model was trained with the specific benchmark, making it an in-domain result.

Model	# Data	In-domain	Out-of-domain				
		SB-Bench	MindCube	MMSI-Bench	SPBench	MMMU	Overall
Proprietary Models							
GPT-5 [19]	-	43.8	56.7	42.8	40.8	77.6	54.5
Gemini-2.5-flash [7]	-	36.4	51.0	32.2	42.6	58.8	46.2
Claude-Sonnet-4.5	-	46.3	34.0	32.7	43.9	69.2	43.9
Open-Source Models							
LLaVA-OneVision-7B [13]	-	23.8	39.0	28.4	43.4	46.7	39.4
InternVL3-2B [29]	-	23.8	33.6	27.5	39.5	41.3	35.5
InternVL3-8B [29]	-	17.2	34.8	29.0	45.6	44.8	38.6
Llama-3.2-11B-Vision-Instruct	-	23.0	29.5	24.5	21.0	40.7	28.9
Spatial Specialist Models							
Qwen2.5-VL-3B Based Models							
Qwen2.5-VL-3B-Instruct [2]	-	29.9	39.1	26.1	<u>41.5</u>	45.6	38.1
SpaceQwen [5]	2B	30.6	38.8	24.5	41.0	45.6	37.5
SpatialMLLM [23]	120K	30.0	36.3	25.4	39.5	40.4	35.4
SpatialLadder-3B [14]	25K	25.5	<u>46.4</u>	26.0	74.0*	47.3	39.9
SpatialBlock-3B-direct (Ours)	15K	94.8	49.1	<u>28.1</u>	39.7	46.4	<u>40.8</u>
SpatialBlock-3B-reason (Ours)	15K	<u>90.2</u>	45.3	29.2	44.6	<u>46.7</u>	41.4
Qwen2.5-VL-7B Based Models							
Qwen2.5-VL-7B-Instruct [2]	-	20.8	31.3	27.4	49.1	46.6	38.6
SpaceR [18]	150K	26.8	31.6	27.4	<u>53.1</u>	<u>55.4</u>	41.9
SpatialBlock-7B-direct (Ours)	15K	95.0	48.8	<u>28.4</u>	48.0	45.2	<u>42.6</u>
SpatialBlock-7B-reason (Ours)	15K	<u>77.5</u>	<u>37.9</u>	29.7	53.7	55.5	44.2
Qwen3-VL-4B Based Models							
Qwen3-VL-4B-Instruct [1]	-	17.0	26.2	27.1	43.4	<u>47.5</u>	36.1
SpatialBlock-4B-direct (Ours)	15K	95.7	51.3	<u>27.8</u>	45.0	47.2	42.8
SpatialBlock-4B-reason (Ours)	15K	<u>94.5</u>	<u>41.1</u>	28.3	<u>44.6</u>	49.6	<u>40.9</u>

Benchmark datasets To evaluate the spatial reasoning capability of SpatialBlock, we conduct experiments on five benchmarks spanning both in-domain and out-of-domain settings. For in-domain evaluation, we introduce SB-Bench, the held-out test split of SpatialBlock-15k, which comprises 600 questions covering the same three task categories. For out-of-domain evaluation, we assess generalization to real-world scenarios using four benchmarks: MindCube [27] for multi-view mental modeling, MMSI-Bench [25] for complex spatial reasoning in diverse scenes, SPBench [14] for single- and multi-image spatial reasoning, and MMMU [28] for general visual perception. Across all benchmarks, we restrict evaluation to multiple-choice questions, as our method is not designed for numerical prediction tasks.

5.2 Main Results

Tab. 1 shows the performance of our method on various spatial reasoning benchmark datasets. Notably, models trained with only 15K synthetic SpatialBlock samples consistently outperform existing spatial specialists on real-scene benchmarks. This result highlights that our block-stacking formulation provides an effective training signal for spatial reasoning.

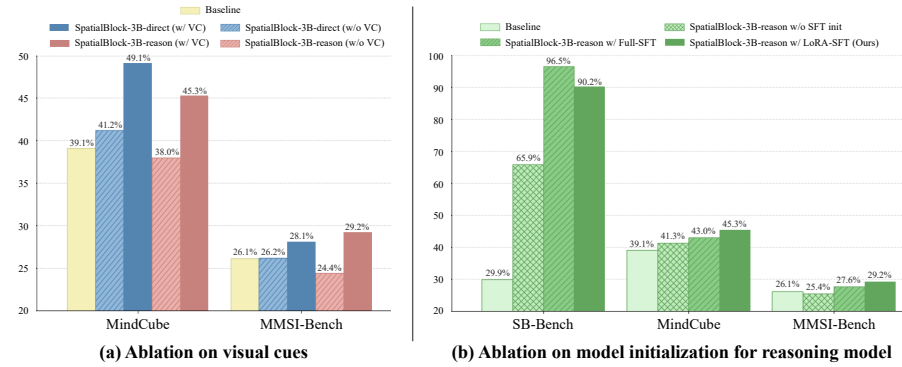


Fig. 4: Ablation Studies. (a) Performance drops significantly without the visual cue in the training dataset. (b) Initializing the model via LoRA-based tuning leads to significant performance gains, whereas both full-parameter tuning and omitting initialization show little to no improvement.

For the **SpatialBlock-direct** models, we observe substantial performance gains, particularly on MindCube, which requires mental simulation. SpatialBlock-3B-direct outperforms the state-of-the-art SpatialLadder by 2.7%. The improvement becomes more pronounced with a bigger backbone: SpatialBlock-7B-direct surpasses its original backbone by 17.6%. Built on the Qwen3 family, SpatialBlock-4B-direct achieves 51.3%, the highest score among open-source models.

The reasoning-enhanced variant, **SpatialBlock-reason**, further demonstrates strong results on MMSI-Bench, which requires complex logical inference. Our SpatialBlock-3B-reason achieves performance comparable to SpatialLadder-3B despite using a much simpler training pipeline based solely on synthetic block-stacking data. SpatialBlock-7B-reason also outperforms SpaceR by 2.3%, even though SpaceR is trained on a dataset ten times larger. In addition, our models consistently improve performance on SPBench, which focuses on relative spatial relations. Although such question types are not explicitly included in our training data, our models generalize better to them than methods trained with explicit supervision.

In summary, each model performance varies by benchmark: the direct model excels on MindCube as it mainly involves canonical 90-degree viewpoint transformations, making direct prediction possible, while the reasoning model achieves stronger results on MMSI-Bench, which requires diverse viewpoint changes and multi-image reasoning. Also, despite being trained without any real-scene images, our models preserve their general visual understanding. Performance on MMMU remains stable, indicating that SpatialBlock training enhances 3D spatial reasoning without degrading broader visual perception.

5.3 Ablation Studies

Ablation on visual cues We introduce color manipulation as a visual cue to better approximate how humans interpret real-world scenes by relying on visually

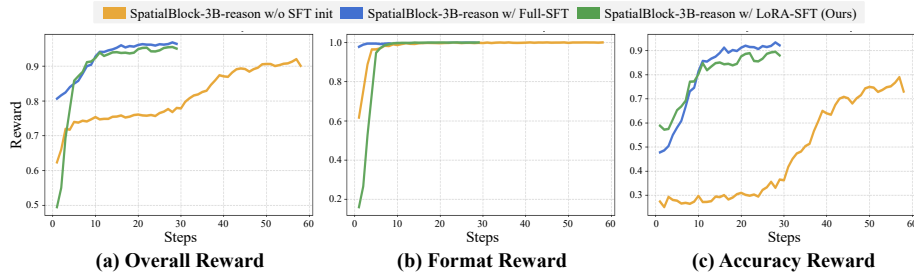


Fig. 5: Training Dynamics of Models with Varying Model Initializations. We compare the reward trajectories across models, including (a) overall, (b) format, and (c) accuracy rewards. To ensure a fair comparison regarding total data exposure, the model without SFT initialization was trained for twice as many iterations.

salient reference objects (Sec. 3.2). To evaluate its effectiveness, we compare models that are trained with and without visual cues. As shown in Fig. 4(a), removing visual cues leads to performance drops in *direct* and *reason* models of 7.9% and 7.3% in MindCube, and 1.9% and 4.8% in MMSI-Bench, respectively. These results suggest that color cues encourage models to better understand the complex scene and mimic human scene perception, thereby improving spatial reasoning in real-world environments.

Ablation on model initialization for reasoning model To train a reasoning-capable model via reinforcement learning, we adopt a LoRA-based initialization strategy to preserve inherent reasoning ability while improving baseline spatial intelligence. Fig. 4(b) compares different initialization design choices and their impact on model performance. Omitting the model fine-tuning stage results in a noticeable performance drop across benchmarks, indicating that this pre-initialization process is necessary to establish basic spatial understanding capabilities. Performing full-parameter fine-tuning with a cold-start phase, commonly used in recent works [9, 14], effectively guides the model to adhere to the required formats and achieves higher accuracy convergence (Fig. 5). However, due to the low quality of reasoning traces used during the cold-start stage (Sec. B.2), it yields only marginal gains. In contrast, LoRA-based initialization provides substantial improvement, establishing a strong starting point for reinforcement learning by endowing the model with both basic spatial intelligence and reliable instruction-following ability.

5.4 Analysis on Spatial Intelligence Improvement

Qualitative analysis Fig. 6 shows reasoning traces of baseline model (Qwen2.5-VL-3B) and models trained on SpatialBlock-15k with and without visual cues for a question from the MindCube dataset. The task requires aggregating visual information from three images captured from different viewpoints. In addition,

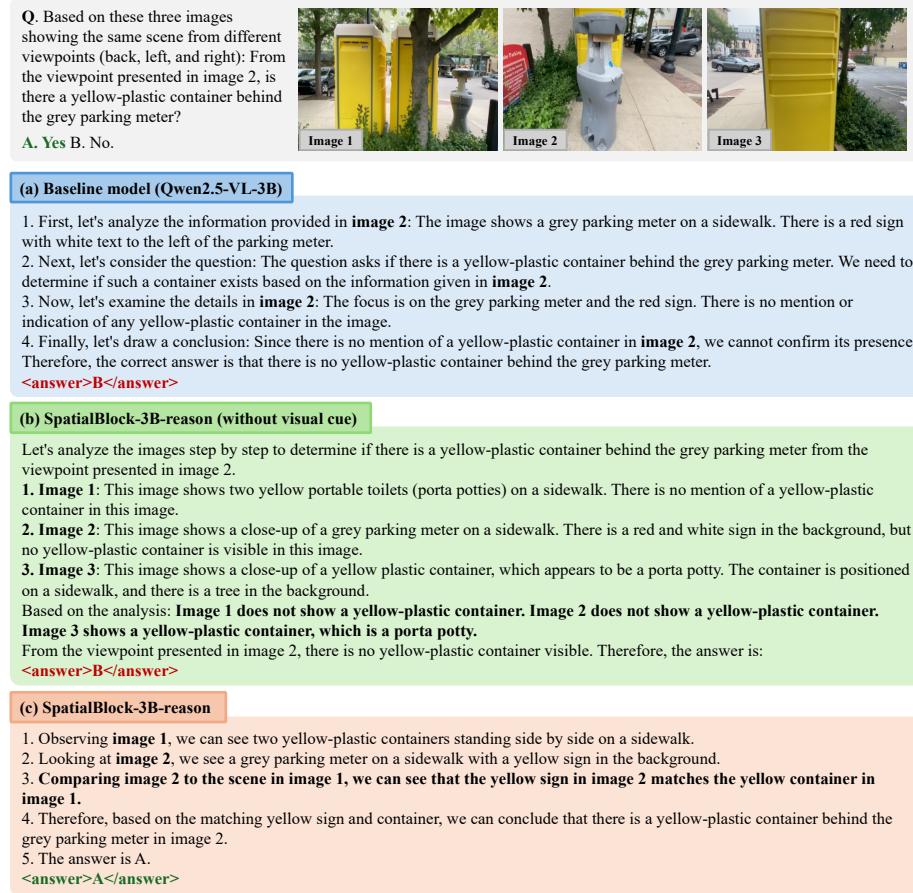


Fig. 6: Qualitative Comparison of Reasoning Traces. (a) Baseline only describes a single view. (b) Training without visual cues attempts multi-image reasoning but fails to identify the shared object. (c) Our model uses the “yellow container” as an anchor to integrate views and reach the correct answer.

the model should infer the spatial relationship between two objects from the perspective presented in the second image (image 2).

As shown in Fig. 6(a), the baseline model fails to exhibit multi-image understanding and merely describes image 2. The model trained without visual cues (Fig. 6(b)) attempts to reason over multiple images, but fails to identify the common object across views, leading to an incorrect conclusion. In contrast, SpatialBlock-3B-reason (Fig. 6(c)) identifies the “yellow container” as an anchor object and connects the given images through this shared reference, ultimately producing the correct answer.

These results demonstrate that training on our dataset enhances the model’s ability to perform complex spatial reasoning. Furthermore, incorporating visual

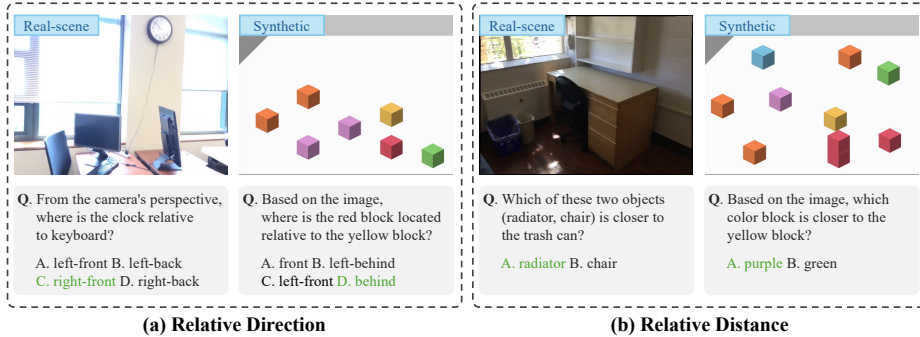


Fig. 7: Synthetic-Real Dataset. Widely used spatial question types, (a) relative direction and (b) relative distance, are adapted into a block-based environment while preserving their semantic structure.

Table 2: Comparison of Training on SpatialBlock and Synthetic-Real. While Synthetic-Real improves in-domain performance, training on SpatialBlock leads to better generalization on out-of-domain benchmark datasets. All experiments are conducted on Qwen2.5-VL-3B [2], and in-domain performance is evaluated on SPBench.

Train Dataset	In-domain		Out-of-domain		
	Relative Direction	Relative Distance	MindCube	MMSI-Bench	MMMU
Baseline	31.4	74.1	39.1	26.1	45.6
Synthetic-Real	43.4	76.9	32.8	25.5	43.6
SpatialBlock	31.4	67.4	49.1	28.1	46.4

cues enables the model to effectively leverage anchor objects to integrate information across multiple viewpoints.

Generalizability of block-stacking questions As shown in Tab. 1, our models trained on SpatialBlock-15k generalize well to real-scene benchmark datasets despite involving question types that differ from those commonly used in existing spatial benchmarks. This observation raises the question of whether our block-stacking tasks provide a more effective learning signal than conventional spatial question formulations. To investigate this, we construct a dataset of the same size as SpatialBlock-15k, termed *Synthetic-Real*, which adapts two widely used spatial question types – relative direction and distance – into our block-based environment (Fig. 7).

Although both datasets can be generated with comparable effort, the results in Tab. 2 reveal a clear difference in outcomes. The model trained on Synthetic-Real achieves higher accuracy on in-domain tasks than the model trained on SpatialBlock. However, it performs poorly on out-of-domain benchmark datasets, even showing performance degradation. In contrast, our model consistently improves across benchmarks. These results suggest that SpatialBlock tasks encourage a more holistic understanding of 3D scenes, whereas conventional spatial

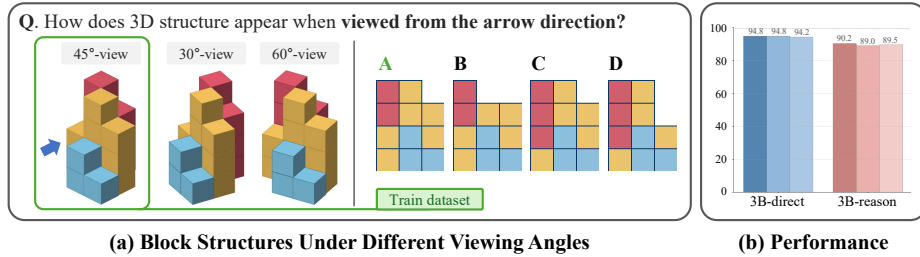


Fig. 8: Performance Across Viewing Angles. (a) The same block structures are rendered from different viewing angles. (b) Performance of models across viewing angles; the bars correspond to results under 45°, 30°, and 60° views, respectively.

questions tend to produce gains that are limited to specific tasks. Consequently, our dataset more effectively promotes general spatial reasoning, enabling more robust and transferable performance.

Robustness to viewing angle variations To further evaluate whether the model learns the underlying 3D structure rather than memorizing appearance patterns from a fixed viewpoint, we conduct a viewpoint generalization experiment. Our SpatialBlock-15k dataset is constructed using blocks rendered from 45° viewing angle, which is fixed during training. We generate the same problems using identical block configurations rendered from different viewing angles (30° and 60°) (Fig. 8(a)). As shown in Fig. 8(b), both the direct answering and reasoning-based models maintain consistent performance across these viewpoints. This suggests that the models capture the underlying structure of the blocks, demonstrating genuine enhancement in spatial ability.

6 Conclusion

In this paper, we revisited the problem of improving spatial intelligence in large vision-language models from a new perspective: learning foundational spatial skills through structured tasks rather than relying on annotation-heavy real-scene supervision. To this end, we introduced **SpatialBlock-15k**, a novel synthetic dataset of block-stacking problems, including 3D-to-2D projection, viewpoint transformation, and structural integration. The dataset further incorporated controlled color variation as visual cues to encourage anchor-based reasoning, mimicking how humans interpret scenes by using specific objects as reference points. Using this dataset, we developed two specialized models: **SpatialBlock-direct** for immediate answer generation and **SpatialBlock-reason** for detailed logical inference. Experimental results demonstrated that our models consistently outperform existing spatial specialists on various real-world spatial benchmarks. Furthermore, extensive analyses show that the learned capabilities generalize across tasks and viewing conditions, indicating that the models capture the underlying spatial structure of scenes. These findings highlighted that robust

spatial intelligence can be achieved through training on carefully designed synthetic spatial tasks, offering a scalable and effective path for improving spatial reasoning in LVLMs.

References

1. Bai, S., Cai, Y., Chen, R., Chen, K., Chen, X., Cheng, Z., Deng, L., Ding, W., Gao, C., Ge, C., et al.: Qwen3-vl technical report. arXiv preprint arXiv:2511.21631 (2025)
2. Bai, S., Chen, K., Liu, X., Wang, J., Ge, W., Song, S., Dang, K., Wang, P., Wang, S., Tang, J., Zhong, H., Zhu, Y., Yang, M., Li, Z., Wan, J., Wang, P., Ding, W., Fu, Z., Xu, Y., Ye, J., Zhang, X., Xie, T., Cheng, Z., Zhang, H., Yang, Z., Xu, H., Lin, J.: Qwen2.5-vl technical report (2025), <https://arxiv.org/abs/2502.13923>
3. Caldera, Y.M., Culp, A.M., O’Brien, M., Truglio, R.T., Alvarez, M., Huston, A.C.: Children’s play preferences, construction play with blocks, and visual-spatial skills: Are they related? *International Journal of Behavioral Development* **23**(4), 855–872 (1999)
4. Casey, B.M., Andrews, N., Schindler, H., Kersh, J.E., Samper, A., Copley, J.: The development of spatial skills through interventions involving block building activities. *Cognition and instruction* **26**(3), 269–309 (2008)
5. Chen, B., Xu, Z., Kirmani, S., Ichter, B., Sadigh, D., Guibas, L., Xia, F.: Spatialvlm: Endowing vision-language models with spatial reasoning capabilities. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. pp. 14455–14465 (2024)
6. Cheng, A.C., Yin, H., Fu, Y., Guo, Q., Yang, R., Kautz, J., Wang, X., Liu, S.: Spatialrgpt: Grounded spatial reasoning in vision-language models. *Advances in Neural Information Processing Systems* **37**, 135062–135093 (2024)
7. Comanici, G., Bieber, E., Schaekermann, M., Pasupat, I., Sachdeva, N., Dhillon, I., Blistein, M., Ram, O., Zhang, D., Rosen, E., et al.: Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. arXiv preprint arXiv:2507.06261 (2025)
8. Feng, Z., Kang, Z., Wang, Q., Du, Z., Yan, J., Shi, S., Yuan, C., Liang, H., Deng, Y., Li, Q., et al.: Seeing across views: Benchmarking spatial reasoning of vision-language models in robotic scenes. arXiv preprint arXiv:2510.19400 (2025)
9. Guo, D., Yang, D., Zhang, H., Song, J., Wang, P., Zhu, Q., Xu, R., Zhang, R., Ma, S., Bi, X., et al.: Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. arXiv preprint arXiv:2501.12948 (2025)
10. Jirout, J.J., Newcombe, N.S.: Building blocks for developing spatial skills: Evidence from a large, representative us sample. *Psychological science* **26**(3), 302–310 (2015)
11. Land, M.F., Hayhoe, M.: In what ways do eye movements contribute to everyday activities? *Vision research* **41**(25-26), 3559–3565 (2001)
12. Levine, S.C., Ratliff, K.R., Huttenlocher, J., Cannon, J.: Early puzzle play: a predictor of preschoolers’ spatial transformation skill. *Developmental psychology* **48**(2), 530 (2012)
13. Li, B., Zhang, Y., Guo, D., Zhang, R., Li, F., Zhang, H., Zhang, K., Zhang, P., Li, Y., Liu, Z., et al.: Llava-onevision: Easy visual task transfer. arXiv preprint arXiv:2408.03326 (2024)
14. Li, H., Li, D., Wang, Z., Yan, Y., Wu, H., Zhang, W., Shen, Y., Lu, W., Xiao, J., Zhuang, Y.: Spatialladder: Progressive training for spatial reasoning in vision-language models. arXiv preprint arXiv:2510.08531 (2025)

15. Lou, H., Fan, C., Liu, Z., Wu, Y., Wang, X.: Llava-sp: Enhancing visual representation with visual spatial tokens for mllms. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 22014–22024 (2025)
16. Ma, W., Chou, Y.C., Liu, Q., Wang, X., de Melo, C., Xie, J., Yuille, A.: Spatial-reasoner: Towards explicit and generalizable 3d spatial reasoning. arXiv preprint arXiv:2504.20024 (2025)
17. Ma, W., Ye, L., de Melo, C.M., Yuille, A., Chen, J.: Spatialllm: A compound 3d-informed design towards spatially-intelligent large multimodal models. In: Proceedings of the Computer Vision and Pattern Recognition Conference. pp. 17249–17260 (2025)
18. Ouyang, K., Liu, Y., Wu, H., Liu, Y., Zhou, H., Zhou, J., Meng, F., Sun, X.: Spacer: Reinforcing mllms in video spatial reasoning. arXiv preprint arXiv:2504.01805 (2025)
19. Singh, A., Fry, A., Perelman, A., Tart, A., Ganesh, A., El-Kishky, A., McLaughlin, A., Low, A., Ostrow, A., Ananthram, A., et al.: Openai gpt-5 system card. arXiv preprint arXiv:2601.03267 (2025)
20. Tian, K., Mao, J., Zhang, Y., Jiang, J., Zhou, Y., Tu, Z.: Nuscenes-spatialqa: A spatial understanding and reasoning benchmark for vision-language models in autonomous driving. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 4567–4576 (2025)
21. Tong, P., Brown, E., Wu, P., Woo, S., Iyer, A.J.V., Akula, S.C., Yang, S., Yang, J., Middepogu, M., Wang, Z., et al.: Cambrian-1: A fully open, vision-centric exploration of multimodal llms. *Advances in Neural Information Processing Systems* **37**, 87310–87356 (2024)
22. Wang, P., Bai, S., Tan, S., Wang, S., Fan, Z., Bai, J., Chen, K., Liu, X., Wang, J., Ge, W., et al.: Qwen2-vl: Enhancing vision-language model’s perception of the world at any resolution. arXiv preprint arXiv:2409.12191 (2024)
23. Wu, D., Liu, F., Hung, Y.H., Duan, Y.: Spatial-mllm: Boosting mllm capabilities in visual-based spatial intelligence. arXiv preprint arXiv:2505.23747 (2025)
24. Yang, J., Yang, S., Gupta, A.W., Han, R., Fei-Fei, L., Xie, S.: Thinking in space: How multimodal large language models see, remember, and recall spaces. In: Proceedings of the Computer Vision and Pattern Recognition Conference. pp. 10632–10643 (2025)
25. Yang, S., Xu, R., Xie, Y., Yang, S., Li, M., Lin, J., Zhu, C., Chen, X., Duan, H., Yue, X., et al.: Mmsi-bench: A benchmark for multi-image spatial intelligence. arXiv preprint arXiv:2505.23764 (2025)
26. Yaowei Zheng, J.L., Shenzhi Wang, Z.F., Dongdong Kuang, Y.X., Zhang, R.: Easyrl: An efficient, scalable, multi-modality rl training framework. <https://github.com/hiyouga/EasyR1> (2025)
27. Yin, B., Wang, Q., Zhang, P., Zhang, J., Wang, K., Wang, Z., Zhang, J., Chandrasegaran, K., Liu, H., Krishna, R., et al.: Spatial mental modeling from limited views. In: Structural Priors for Vision Workshop at ICCV’25 (2025)
28. Yue, X., Ni, Y., Zhang, K., Zheng, T., Liu, R., Zhang, G., Stevens, S., Jiang, D., Ren, W., Sun, Y., et al.: Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 9556–9567 (2024)
29. Zhu, J., Wang, W., Chen, Z., Liu, Z., Ye, S., Gu, L., Tian, H., Duan, Y., Su, W., Shao, J., et al.: Internvl3: Exploring advanced training and test-time recipes for open-source multimodal models. arXiv preprint arXiv:2504.10479 (2025)

A Details on Construction and Evaluation of SpatialBlock-15k

A.1 Data Construction Process

All 3D structures in SpatialBlock-15k are constructed on a 3×3 spatial grid, where each grid position can contain between one and four vertically stacked blocks. The height of each stack defines the underlying 3D configuration. To avoid ambiguity when interpreting rendered structures, we restrict the set of valid configurations so that the number of blocks at every grid position can be reliably inferred from the rendered image. In particular, configurations that have severe occlusion – where blocks are completely hidden behind others from the viewing perspective – are excluded. This constraint ensures that the global structure remains visually interpretable while still allowing a diverse set of configurations.

Based on the resulting valid structures, we generate the three types of tasks illustrated in Fig. 3. For each task, the distractor choices are carefully constructed to be structurally plausible while incorrect, preventing simple heuristics and encouraging models to reason about the spatial relationships within the structure.

Q1: 3D-to-2D projection with depth cues In this task, the model must predict the correct 2D projection (e.g., front or side view) of a given 3D block structure. The wrong choices are generated through controlled perturbations of the correct projection:

- **Block-removal perturbation.** A visible top block from one column is removed, producing a projection that corresponds to a slightly reduced stack height while remaining structurally plausible.
- **Color-consistency perturbation.** For a column containing multiple colors, the color of one block is replaced with that of a neighboring block in the same column. This preserves the geometric structure but violates the correct color arrangement.
- **Combined perturbation.** A hybrid modification that simultaneously removes a block from one column and alters the color assignment in another column.

During choice generation, perturbations are only applied to columns that contain visible blocks and sufficient color diversity, ensuring that distractors remain meaningful and visually distinguishable.

Q2: Viewpoint transformation with anchor blocks This task evaluates whether models can reason about global spatial transformations of the structure. Given an original structure, one transformation is randomly selected from the following set: 90° , 180° , or 270° rotations, and flipping vertically. A specific visible block in the original structure is highlighted, and the model needs to identify the corresponding block position after the transformation. To ensure

valid supervision signals, only blocks that remain visible after transformation are selected as colored blocks.

The answer choices consist of four candidates:

- the correctly transformed structure with the correct mapped block position (correct answer),
- the original structure with a different block highlighted,
- the correctly transformed structure with an incorrectly highlighted block,
- or a structure generated through a different transformation.

Q3: Structural combination with overlapping blocks This task focuses on reasoning the union when two block structures are combined. Each problem instance consists of two structures and a specified attachment point indicated by overlapping color. The goal is to determine the correct outcome of the combination. During dataset construction, we sample a possible combination from the given 3D structures and compute their first contact location as the overlap region.

The distractor choices are constructed by choosing a 3D configuration where one component is replaced, and the other remains unchanged, resulting in a different structure. By ensuring that each distractor differs only in one of the two input structures, the choices remain visually similar to the correct answer while requiring the model to reason about the precise spatial interaction between the specific pair of structures.

A.2 Human Evaluation on SB-Bench

For human evaluation, we randomly select 30 samples from SB-Bench and collect responses from 31 participants. As illustrated in Fig. 1, humans achieve 95% accuracy on average, demonstrating the task’s simplicity.

B Additional Training Details and Results

B.1 Training Implementation Details

Early stopping for SpatialBlock-direct To prevent overfitting to the block structure during direct model training, we selected the model from an intermediate iteration in which test accuracy exceeded 93%, specifically the iteration immediately before it began to decrease.

Reasoning prompt used for reinforcement learning The detailed instructions for reinforcement learning are shown in Fig. 9. The prompt guides the model to output reasoning steps in a numbered sequence and gives the final answer between `<answer>` and `</answer>` tags. We do not use the extra `<reason>` tag to reduce the load of learning the format.

RL prompt

Please think about this question step by step and provide your detailed reasoning in a numbered reasoning sequence (1., 2., 3., ...). Then give your final answer between the `<answer>` `</answer>` tags. Output must be EXACTLY in this format (including tags):

...

`<answer>`X`</answer>`

Please provide only the single option letter (e.g., A, B, C, D, etc.) within the `<answer>` `</answer>` tags.

Fig. 9: Instructions for Reinforcement Learning. The model is instructed to provide a step-by-step thinking process, and then the final answer between answer tags.

Hyperparameters used for training We apply LoRA-based supervised fine-tuning for SpatialBlock-reason, using rank 8 for SpatialBlock-3B model and rank 16 for SpatialBlock-7B and 4B models. For reinforcement learning, we use EasyR1 [26] as our base code and follow its default hyperparameters. We set the batch size to 512 with 5 rollouts, while extending the maximum prompt length to 4096 to accommodate multi-image training. All training was conducted for 1 epoch using SpatialBlock-15k.

B.2 Additional Results on SpatialBlock-reason Initialization

Cold-start data quality As mentioned in Sec. 5.3, full-parameter fine-tuning with cold-start phase shows marginal improvement compared with LoRA-based initialization. We hypothesize that it is due to the low quality of the cold-start reasoning trace. Fig. 10 shows a reasoning example extracted from Qwen2.5-VL-7B. Although we choose the one with correct answers, the model fails to explain the 3D structure correctly, particularly miscounting the blocks. Furthermore,

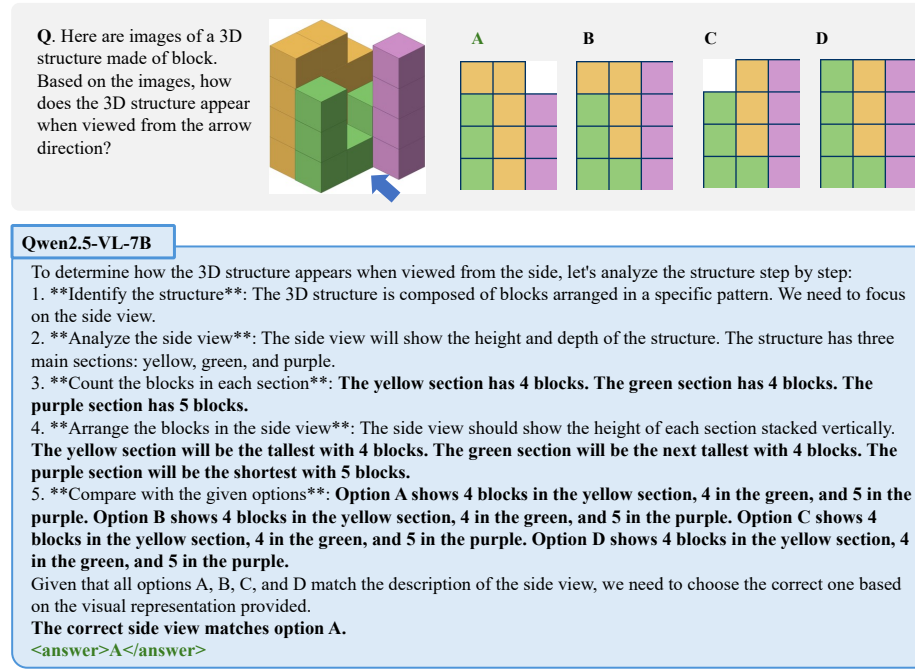


Fig. 10: Generated Cold-Start Data Sample. The reasoning trace extracted from Qwen2.5-VL-7B includes incorrect information about the given 3D structure, even though the model answers correctly.

despite the attempt to evaluate all options, it generates the same explanation for distinct options, leading to a random final choice.

We also observed that even propriety model such as GPT-5 frequently generates inaccurate block descriptions and reasoning, highlighting the difficulty of producing stable trajectories even for top-tier models on our tasks. Using these GPT-5-generated labels for cold-start SFT still underperforms ours especially on spatial reasoning benchmarks (Tab. 3), which supports our claim that low-quality teacher reasoning can serve as noisy supervision.

LoRA-initialized model performance We choose LoRA-based SFT as the initialization strategy of SpatialBlock-reason rather than tuning full parameters, as the latter yields only marginal improvement even with the cold start process (Sec. 5.3). As detailed in Tab. 3, LoRA-SFT improves MindCube by over 6% across all models, but Qwen2.5-VL-based models show negligible gains or slight declines in other out-of-domain datasets. However, when combined with RL, the model achieves significant performance gains across out-of-domain benchmarks, demonstrating enhanced spatial reasoning and robust generalization on visual understanding tasks such as MMMU.

Table 3: Performance Regarding SpatialBlock-reason Initialization. The table shows the performance of intermediate LoRA-tuned models of SpatialBlock-reason and GPT-5 coldstart initialized SpatialBlock-reason.

Model	Training Strategy	In-domain	Out-of-domain			
		SB-Bench	MindCube	MMSI-Bench	SPBench	MMMU
Qwen2.5-VL-3B-Instruct [2]	Baseline	29.9	39.1	26.1	41.5	45.6
SpatialBlock-3B-reason (ours)	LoRA-SFT	92.2	46.6	25.1	40.2	46.6
	+RL	90.2	45.3	29.2	44.6	46.7
SpatialBlock-3B-reason (with GPT-5 cold-start)	cold-start SFT + RL	75.3	41.7	28.0	46.3	48.1
Qwen2.5-VL-7B-Instruct [2]	Baseline	20.8	31.3	27.4	49.1	46.6
SpatialBlock-7B-reason (ours)	LoRA-SFT	74.2	37.8	28.1	46.3	45.5
	+RL	77.5	37.9	29.7	53.7	55.5
Qwen3-VL-4B-Instruct [1]	Baseline	17.0	26.2	27.1	43.4	47.5
SpatialBlock-4B-reason (ours)	LoRA-SFT	96.5	36.0	28.5	44.5	51.5
	+RL	94.5	41.1	28.3	44.6	49.6

B.3 Results on Additional Task and Model

Table 4: Results on Numerical Task and InternVL3

(a) Results on Numerical Tasks. Results on tasks from SPBench.

Model	Absolute distance	Object size	Object counting
Qwen2.5-VL-3B-Instruct	30.9	17.6	35.1
SpatialBlock-3B-direct	23.2	19.9	72.1
SpatialBlock-3B-reason	21.4	38.3	43.3

(b) Results on InternVL3 backbone.

Model	In-domain	Out-of-domain			
	SBench	MindCube	MMSI-Bench	SPBench	MMMU
InternVL3-2B [29]	19.0	32.1	27.0	33.0	41.3
SpatialBlock-2B-direct	97.2	51.4	27.3	41.3	42.4
SpatialBlock-2B-reason	78.3	38.9	27.8	45.6	47.3

Numerical Task Result We evaluate our model on numerical prediction tasks from SPBench (Tab. 4a). Despite being trained only on multiple-choice questions, our model improves in counting and size estimation, as our dataset enhances the model’s ability to individuate objects and infer their scales. In contrast, it is less effective at absolute distance because our method prioritizes structural understanding and spatial integration over precise metric regression.

Other Model Family Result We conduct training on InternVL3-2B [29], a widely used post-training backbone. As shown in Tab. 4b, our method consistently improves InternVL3, demonstrating generalization beyond the Qwen family.

B.4 Additional Ablations Studies

Ablation on Question Types We additionally evaluate direct models trained on a single question type (Tab. 5(a)). Q1-only shows slight gains on MindCube/MMSI by focusing on viewpoint changes, while Q2-only excels on MindCube due to its 90-degree transformations. Q3-only yields incremental gains across all benchmarks by fostering holistic scene integration from multiple perspectives. Individually, however, these gains remain inconsistent. In contrast,

Table 5: Additional Ablation Studies.

Model	In-domain	Out-of-domain			
	SBBench	MindCube	MMSI-Bench	SPBench	MMMU
<i>(a) Ablation on question types</i>					
Q1 only	47.0	43.7	28.7	41.5	43.9
Q2 only	53.8	48.5	27.1	38.9	47.8
Q3 only	47.5	42.3	28.7	46.1	45.5
<i>(b) Train with SpatialLadder-26k (*in-domain)</i>					
SpatialBlock-3B-direct (SL)	24.7	44.5	27.2	72.1*	45.3
SpatialBlock-3B-reason (SL)	28.0	44.9	27.5	70.0*	44.7

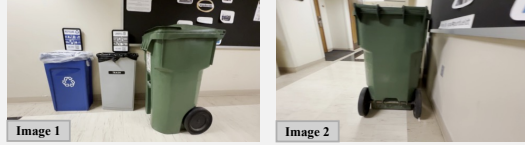
training on all three types achieves the most robust performance, confirming that our question types provide complementary spatial skills necessary for generalized OOD reasoning.

Ablation on Task Design In order to analyze the effect of the task design itself more thoroughly, we train a model on SpatialLadder-26k using the identical training recipe. For fair comparison, we use 11.5k multiple-choice questions matching our response format and train for 1.3 epochs to match the same training size. As shown in Tab. 5(b), this model underperforms ours on most benchmarks except the in-domain benchmark SPBench, suggesting that our performance gains mainly stem from the proposed block-stacking task design.

B.5 Additional Qualitative Result

Q. Based on these two views showing the same scene: in which direction did I move from the first view to the second view?

- A. Directly right B. **Diagonally forward and right**
C. Directly left D. Diagonally forward and left



(a) Baseline model (Qwen2.5-VL-3B)

To determine the direction of movement from the first view to the second view, let's analyze the given information:

- First View**: The image shows three trash bins arranged side by side. From left to right, they are: Blue recycling bin, Gray trash bin, Green trash bin with wheels.
- Second View**: The image shows the green trash bin with wheels positioned against a wall. Given this information, we can deduce the following: The blue recycling bin is on the far left. The gray trash bin is in the middle. The green trash bin with wheels is on the far right. In the second view, the green trash bin with wheels is positioned against a wall, **indicating that it has been moved from its original position next to the other bins to a new location.**

Now, let's consider the possible directions of movement:

- Directly Right**: This would mean **moving the green trash bin from its original position to the right**, which aligns with the description of it being positioned against a wall.
- Diagonally Forward and Right**: This would imply moving the green trash bin diagonally from its original position to the right, which does not match the description.
- Directly Left**: This would mean moving the green trash bin from its original position to the left, which does not match the description.
- Diagonally Forward and Left**: This would imply moving the green trash bin diagonally from its original position to the left, which does not match the description.

Based on the analysis, the most logical direction of movement is directly right.
Therefore, the correct answer is:
<answer>A</answer>

(b) SpatialBlock-3B-reason (without visual cue)

Let's analyze the two views step by step:

- View 1**: We see three trash bins: a blue recycling bin, a gray trash bin, and a green trash bin. There is also a blackboard with some stickers on it.
- View 2**: **The green trash bin is now in a different position, suggesting that it has been moved.** The other two bins (blue recycling bin and gray trash bin) are still in the same position, **but the green trash bin is now closer to the camera.**
- Direction Analysis**: From View 1, the green trash bin is to the right of the blue recycling bin and the gray trash bin. In View 2, the green trash bin is still to the right of the blue recycling bin and the gray trash bin, but it is now closer to the camera, indicating a movement in the direction of the camera.

Based on this analysis, the direction of movement from View 1 to View 2 is diagonally forward and right.
<answer>B</answer>

(c) SpatialBlock-3B-reason

- Observing view 1, we see three trash bins: a blue recycling bin, a gray trash bin, and a green trash bin with wheels.
- In view 2, **the green trash bin with wheels is the only one visible**, and it appears to be in a different position.
- The green trash bin with wheels in view 2 is positioned closer to the wall and appears to be in a different orientation compared to view 1.
- Based on the change in the position and orientation of the green trash bin, it seems that the camera has moved diagonally forward and to the right.
- Therefore, the direction from view 1 to view 2 is diagonally forward and right.

<answer>B</answer>

Fig. 11: Additional Qualitative Result on Reasoning Traces. The figure shows additional qualitative results on reasoning traces of baseline model, our model trained without visual cue, and our final model, SpatialBlock-3B-reason.

We illustrate additional reasoning traces from the baseline model and the models trained with our dataset variants. Fig. 11 shows that the baseline model

and the model trained without visual cues incorrectly interpret a viewpoint change as the movement of a certain object. Furthermore, while the baseline model verbosely explains the scene, our model trained with SpatialBlock-15k provides a concise explanation based on the anchor object (e.g., green trash bin) and derives its reasoning correctly.

To further assess reasoning quality, we utilize MMSI-Bench with its ground-truth rationales. GPT-5 scores each reasoning’s alignment with each ground-truth step as 1 (present), 0.5 (implicit), or 0 (incorrect), and the final score is averaged and scaled to 100. Our model achieves an average score of 21.1, compared to 17.8 for the baseline, indicating consistently improved reasoning quality beyond better formatting or instruction following.